The Dual Labour Market and the Motherhood Employment Penalty in Japan

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The Puzzle

Japan's childcare leave can be extended up to 24 months under eligibility conditions, and Japan has one of the longest father-specific entitlements in the OECD in *full-rate equivalent* terms (about 31 weeks) (OECD, 2024c).

Yet maternal employment when the youngest child is aged 0–2 remains below the OECD average in the latest OECD comparison (OECD, 2024a).

If the policy infrastructure exists, why do women still leave?

In this paper, I show that the key margin is job type: formal eligibility exists, but continuity through childbirth is much weaker in non-regular tracks.

Regular = open-ended, internal career track jobs; non-regular = fixed-term/part-time/dispatch jobs.

What This Paper Is and Isn't

What it is:

- Descriptive event study of 662 first births (KHPS/JHPS, 2004–2022)
- Predictive risk stratification: which jobs predict exit?
- Transparent about what is robust, what is fragile, and what is interpretive

What it is not:

- Not a causal identification design
- Not administrative data (household panel, N=662)
- Not nationally weighted (sample-average trajectories)

If you take one thing away: the penalty is concentrated in jobs with weaker contractual protections.

Japan's Dual Labour Market

Regular (seishain)

- Open-ended contracts
- Seniority wages, firm training
- Strong de facto job protection
- Leave is exercisable

Non-regular (hi-seiki)

- Fixed-term, part-time, dispatch
- Flat wages, limited progression
- Contracts can lapse at/around leave
- Leave is formally available but fragile

- Over 50% of employed women hold non-regular contracts (OECD, 2024b)
- Small firms face tighter staffing and higher adjustment costs around leave (OECD, 2024b)
- Secondary-earner tax/social-insurance thresholds (e.g., 103/130 man-yen) shape hours and re-entry incentives (OECD, 2024b; Nagase, 2012)
- Men's long working hours remain common (e.g., a sizeable share work >48 hours/week) (OECD, 2024b)

Data and Sample I

Data: Harmonised KHPS/JHPS household panels, 2004–2022.

Sample: 1,184 first births $\rightarrow$ 662 in event window $\rightarrow$ 5,212 person-years

- KHPS (since 2004) and JHPS (since 2009) are nationally fielded household panels with harmonised core labour variables; I pool them using the official harmonisation framework and verify robustness to survey source (KHPS vs JHPS split).
- Panel support is unbalanced over event time (women observed: 186 at $t = -5$, 662 at $t = 0$, 454 at $t = +5$).

Data and Sample II

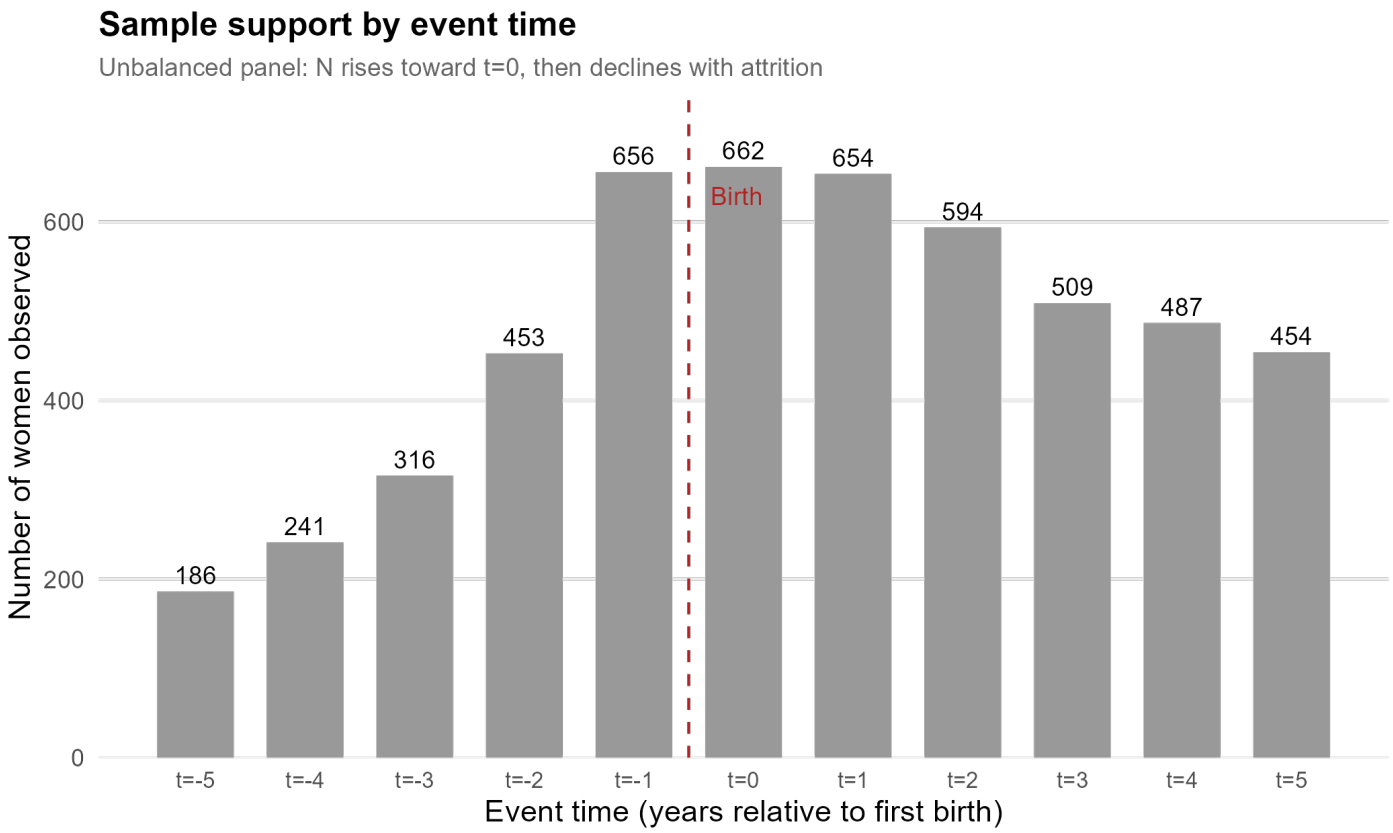


Figure: Event-time support in the unbalanced panel. Support peaks around childbirth and thins in far leads/lags, motivating primary interpretation in $[-2, +2]$.

Specification and Estimand

Specification:

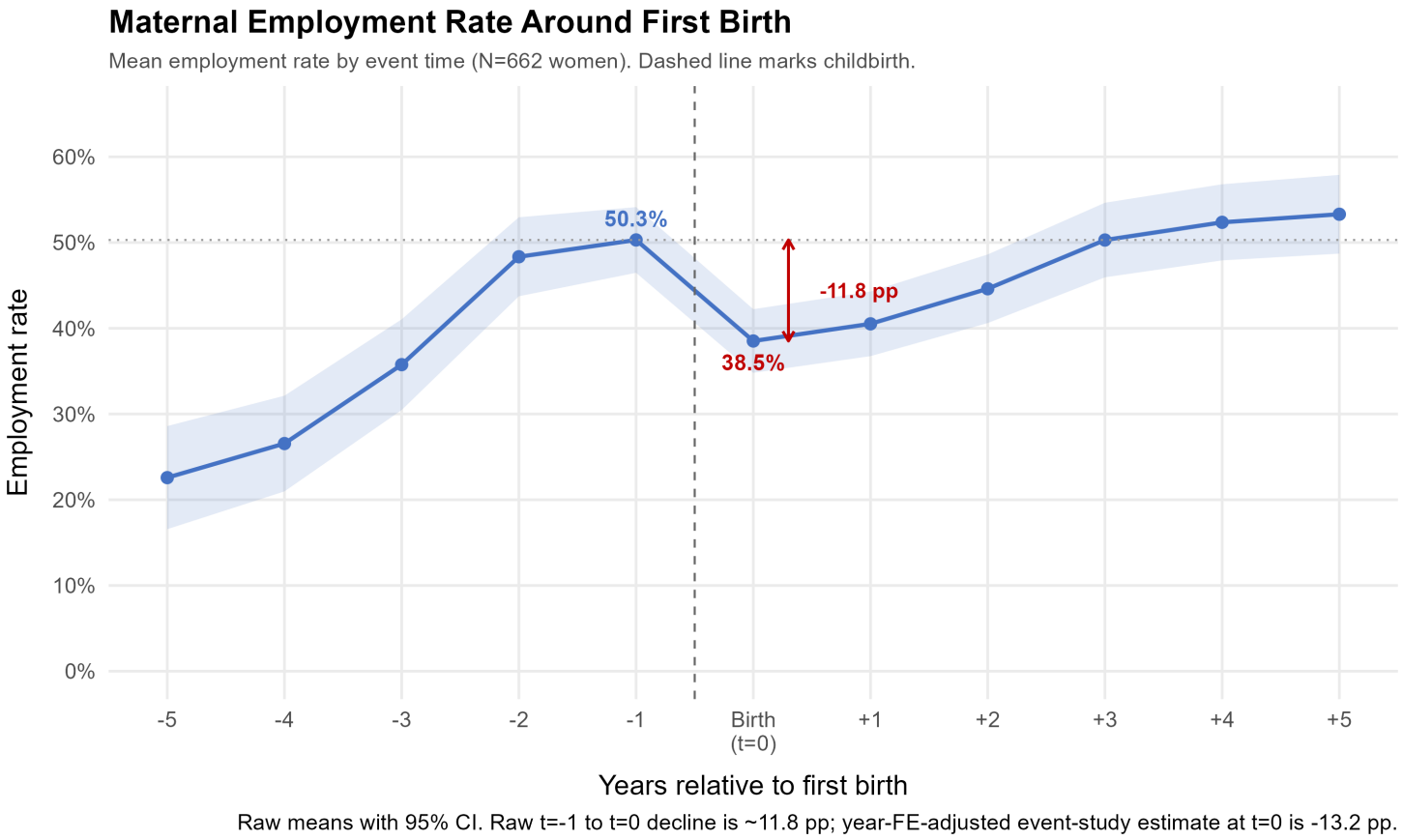
$$Y_{it} = \sum_{k \neq -1} \beta_k \mathbf{1}\{\text{event_time}_{it} = k\} + \gamma_t + \varepsilon_{it}$$

- Y_{it}: outcome for woman i in survey year t; β_k: event-time coefficients relative to $t = -1$; γ_t: calendar-year fixed effects.
- Calendar-year FE, heteroskedasticity-robust SEs; woman-clustered SEs give the same qualitative inference.
- No individual FE in the main specification: individual FE + year FE = collinearity and inflated SEs in this short unbalanced panel. Point estimates are similar, but inference becomes uninformative.

Estimand: Descriptive sample-average event-time trajectories in this analytic panel (not within-person causal treatment effects).

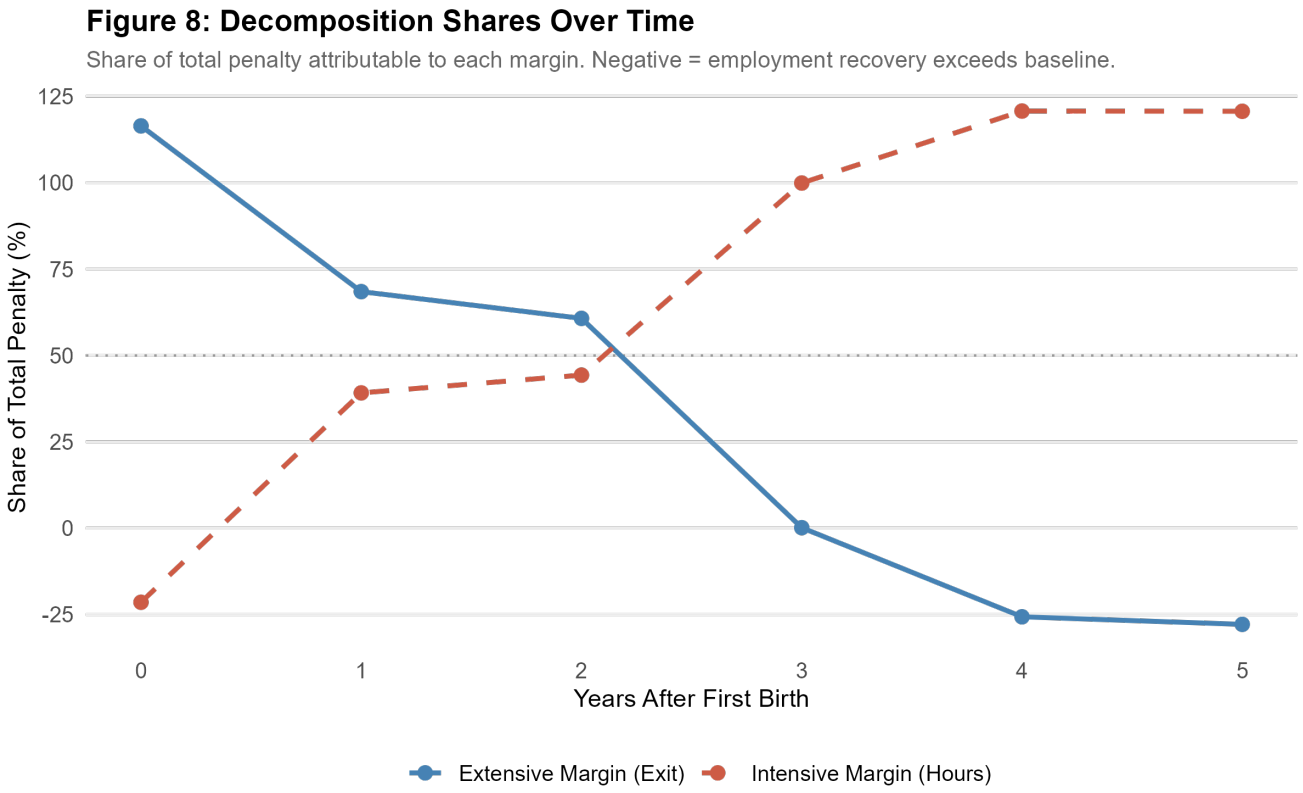
Interpretation strategy: Focus on $[-2, +2]$, where support is strongest; treat far leads/lags as composition-sensitive diagnostics.

Result 1: The Birth-Year Break



Raw employment levels shown; the year-FE-adjusted drop at $t = 0$ is -13.2 pp. Employment is flat at $t = -2$. Then it drops from 50.3% to 38.5% at birth. The pre-birth rise is life-cycle entry, not a pre-trend. The far leads are composition, not anticipation.

But the Recovery Is an Illusion



The composition of the penalty flips over time: the extensive margin (employment exit) explains most of the gap right after birth, but by $t = +3$ the intensive margin (hours/wage losses among returners) becomes dominant and remains so through $t = +5$.

Result 2a: Who Exits at Childbirth? (Risk Profile)

Risk set: women employed at $t = -1$, observed at $t = 0$ (N=330; 124 exits).

Short answer: women in non-regular jobs.

- Exit risk is much lower in protected jobs and much higher in precarious jobs.
- Regular workers in large firms: **13.8%** exit.
- Non-regular workers in small firms: **64.8%** exit.
- Household-side variables add little once job type is included.

Result 2b: Model Summary and Robustness

Model summary

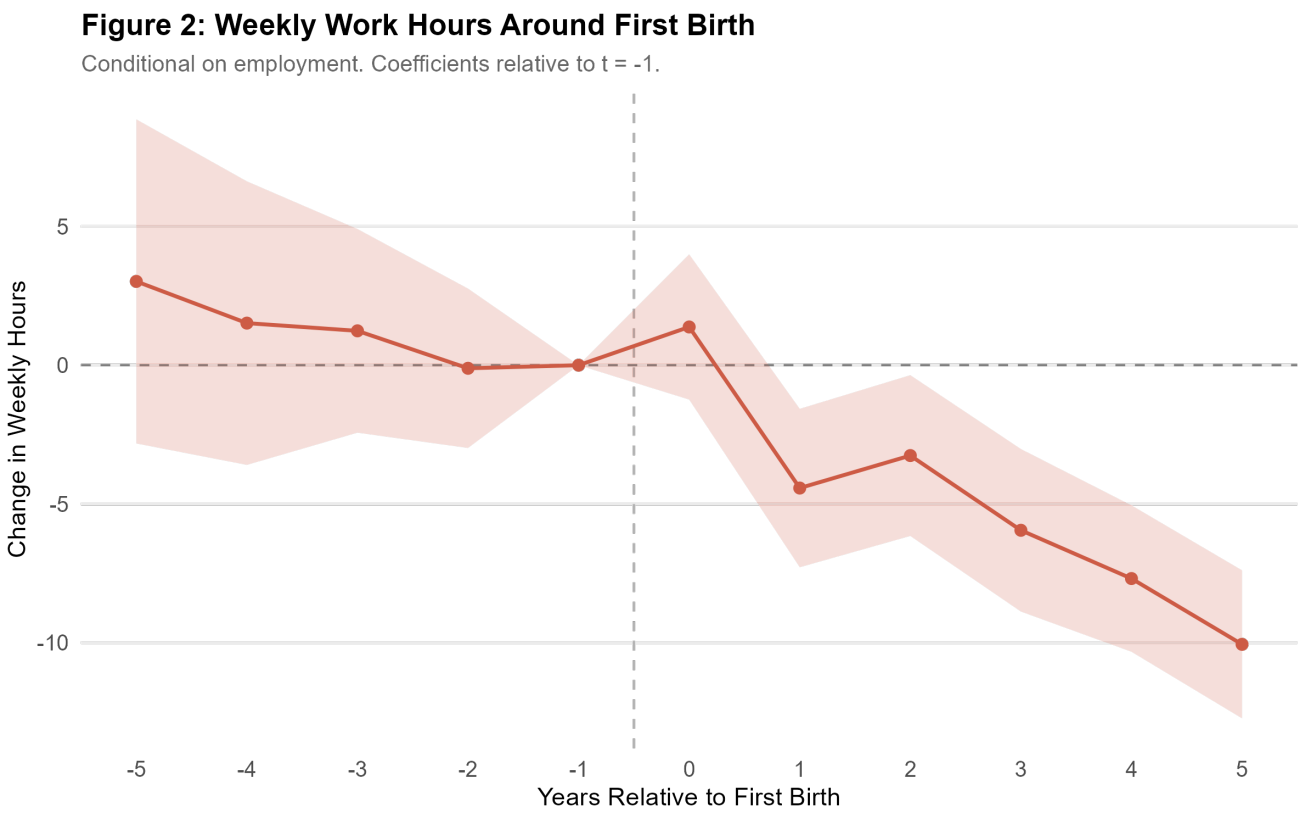
- Non-regular status: strong predictor of childbirth-margin exit (OR $= 7.50$, $p < 0.001$).
- Small-firm effect: positive but less precise (OR $= 1.55$, $p = 0.126$).

Robustness to missing-data handling

- Complete-case OR: 7.50
- Missing-indicator OR: 7.48
- High-information OR: 7.39

Takeaway: the non-regular gradient is stable across specifications, so this is not a missing-data artifact.

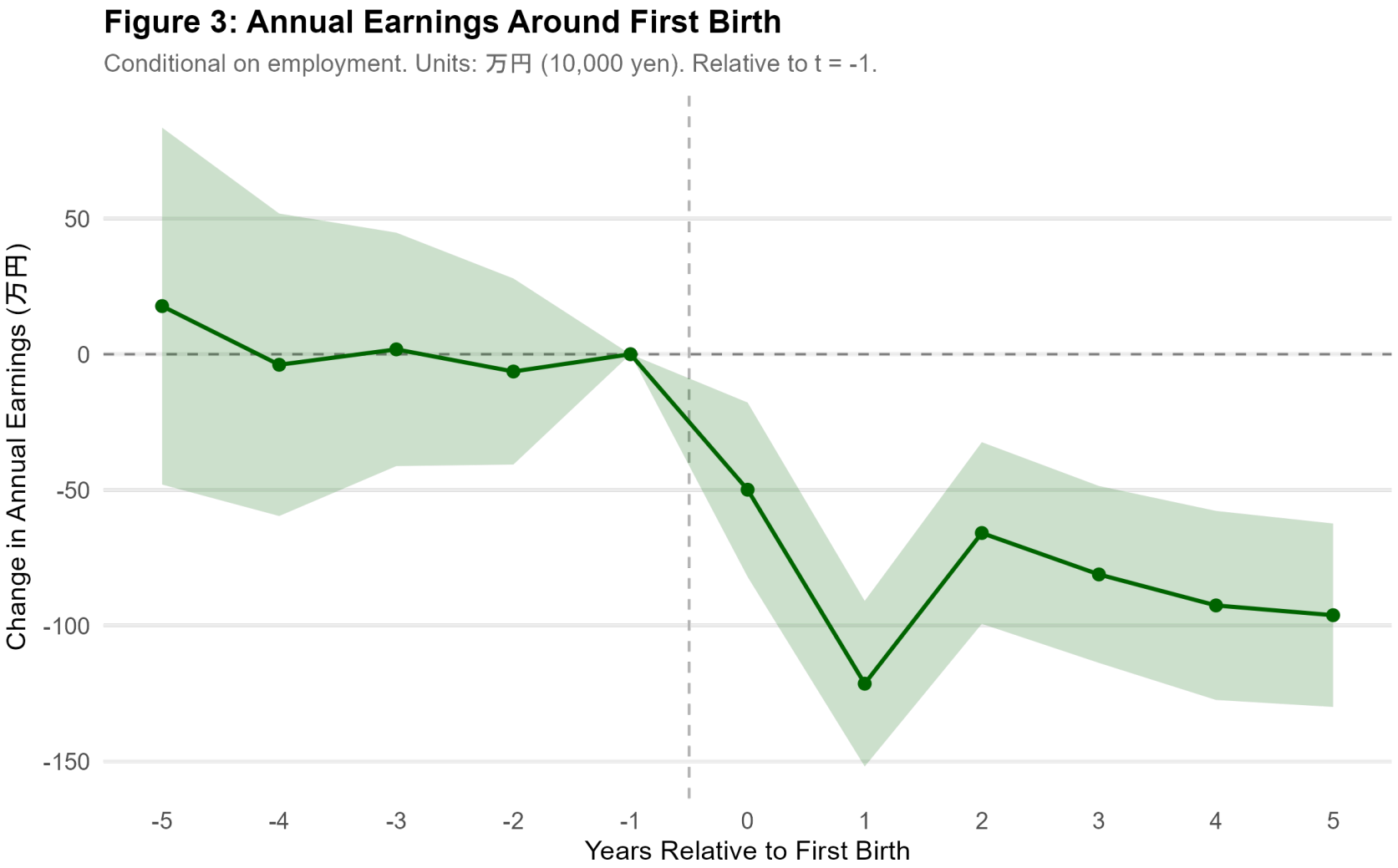
Result 3a: Persistent Hours Losses



Conditional on employment (relative to $t = -1$):

- Coefficients are changes in weekly hours versus the pre-birth reference year.
- By $t = +5$: -10.1 hours/week, about a 30% decline from the $t = -1$ baseline of 34.1.

Result 3b: Persistent Earnings Losses I



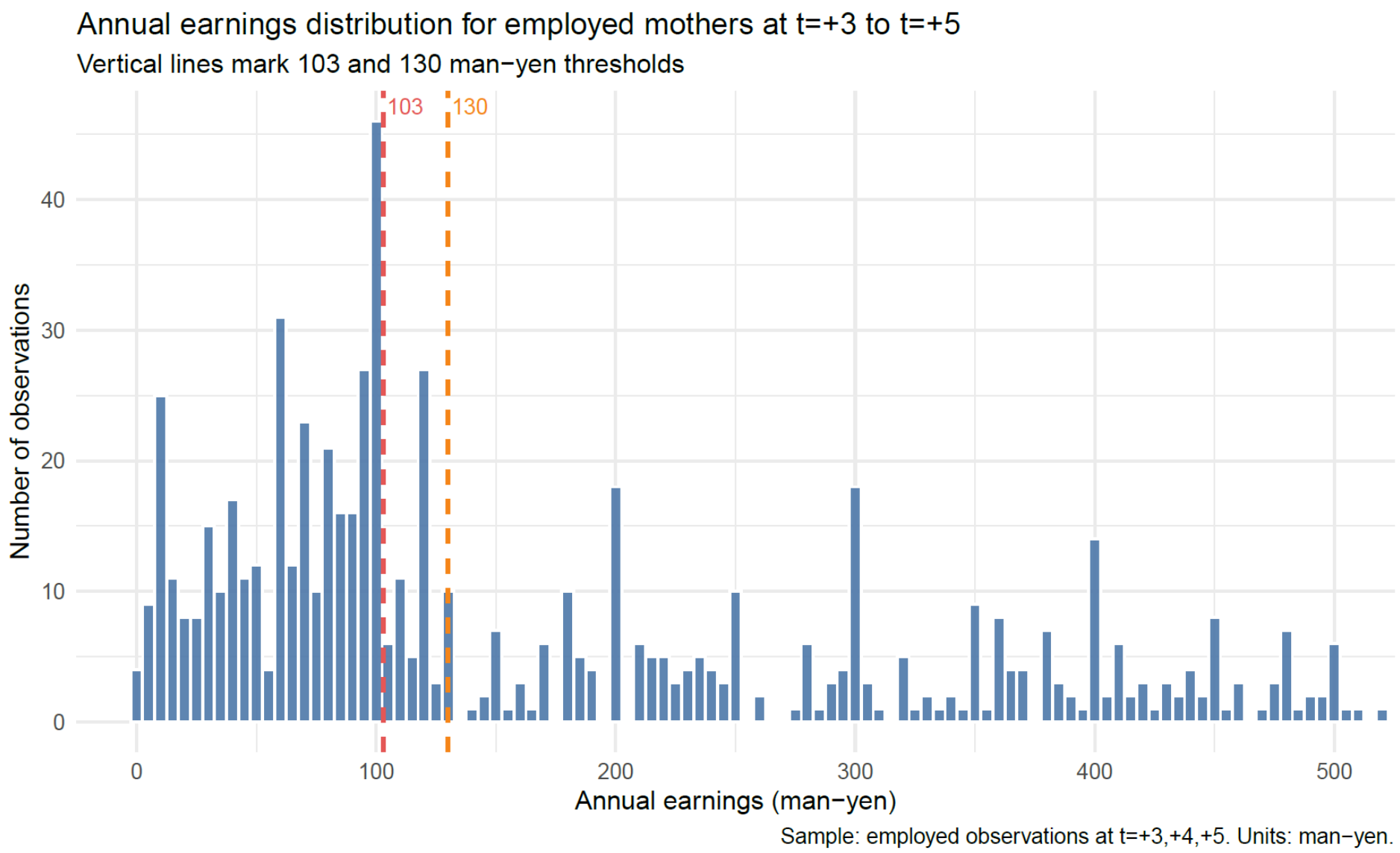
Result 3b: Persistent Earnings Losses II

Conditional on employment (relative to $t = -1$):

- Earnings fall more than hours and do not recover within five years.
- By $t = +5$: -96.2 man-yen (about a 36% decline from baseline 268.9; implied level ≈ 172.7).
- Implied hourly wage also declines (-7.1%), consistent with lower-quality re-entry.

Unconditional outcomes (coding 0 when non-employed) show the same qualitative pattern.

Result 4: Earnings Bunching at Tax Thresholds I



Result 4: Earnings Bunching at Tax Thresholds II

- Among employed mothers at $t = +3$ to $t = +5$, **47.8%** earn ≤ 103 man-yen.
- In the same window, **55.9%** earn ≤ 130 man-yen.
- These thresholds are salient secondary-earner tax/social-insurance benchmarks.
- Net-cost jumps around these points can reduce incentives to increase hours, reinforcing low-hours re-entry.

Notes: Shares are my calculations from KHPS/JHPS returners at $t = +3$ to $t = +5$. Threshold definitions follow OECD (2024b) and Nagase (2012).

Evidence Hierarchy: What Is Solid vs Cautious

Most credible

- Sharp childbirth break in employment at $t = 0$ (-13.2pp).
- Near lead at $t = -2$ is close to zero.
- Childbirth-margin exit risk is strongly stratified by non-regular status (OR ≈ 7.5).

Credible but more sample-sensitive

- Long-horizon ($t = +5$) levels and recovery magnitudes.
- Small-firm effects conditional on contract type.

Interpretive context (not separately identified)

- Childcare logistics, paternal-hours narrative, and institutional mechanisms.

Full stability map and all diagnostics are in backup (placebo, IPW, balanced panel, leave recoding, missingness sensitivity, clustered SE, cohort split, Lee-style trimming).

Implications

The penalty begins as exit and transforms into permanent downgrade. Policy implications consistent with the descriptive evidence:

1. **Protect non-regular workers** - ensure leave rights survive contract renewal
2. **Support small firms** - replacement-hiring subsidies, simplified leave admin
3. **Address men's overwork** - near-zero paternal adjustment in this sample
4. **Break the re-entry trap** - training subsidies are invisible ($<3\%$ usage); fiscal thresholds cap earnings

Bottom line: Expanding childcare and leave helps workers who already hold secure jobs. The binding constraint is labour-market structure - the divide between jobs that can accommodate parenthood and jobs that cannot.

Thank you

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Paper, code, and all robustness tables available on request.

References I

Nagase, N. (2012). Japanese female labor supply and the tax/social insurance system. *Japanese Journal of Labour Studies*, 54(12):12–25.

OECD (2024a). Maternal employment rates (LMF1.2). OECD Family Database. Data for 2021 or latest available year. Accessed 2026-02-25.

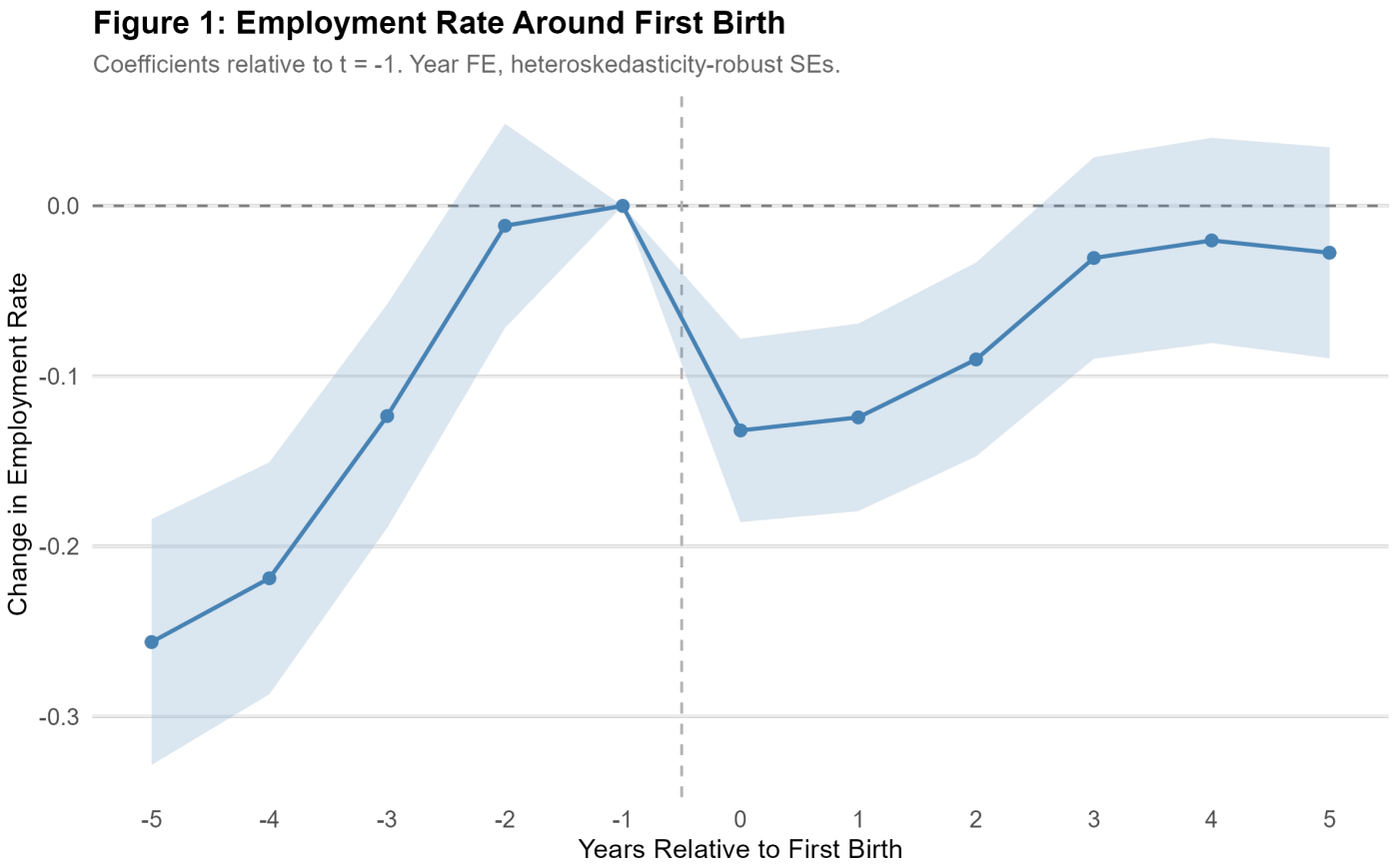
OECD (2024b). OECD economic surveys: Japan 2024. Technical report, Paris.

OECD (2024c). Parental leave systems (PF2.1). OECD Family Database. Data as of April 2024. Accessed 2026-02-25.

Backup: Summary Statistics at $t = -1$

	Full Sample (N=662)	Quitters (N=29)	Stayers (N=262)
Employed (%)	50.3	55.2	61.0
Full-time (%)	30.9	31.0	36.7
Part-time (%)	19.4	24.1	24.3
Hours/week (mean)	34.1	32.5	33.4
Income, man-yen (mean)	268.9	148.4	302.7

Backup: Coefficient Event Study



Standard event-study coefficients relative to $t = -1$. Year FE, robust SEs.

Backup: Pre-Trend Diagnostics

The near lead at $t = -2$ is null across four specifications:

- Full sample: -0.012, $p = 0.70$
- Trimmed pre-period (drop $t = -5, -4, -3$): $p = 0.72$
- Balanced pre-support: $p = 0.078$
- With KHPS/JHPS dataset indicator: $p = 0.71$

Matched childless placebo: $t = 0$ coefficient $= +0.016$, $p = 0.608$.
Joint far-lead test: $p = 0.749$.

Backup: Cohort Split and COVID

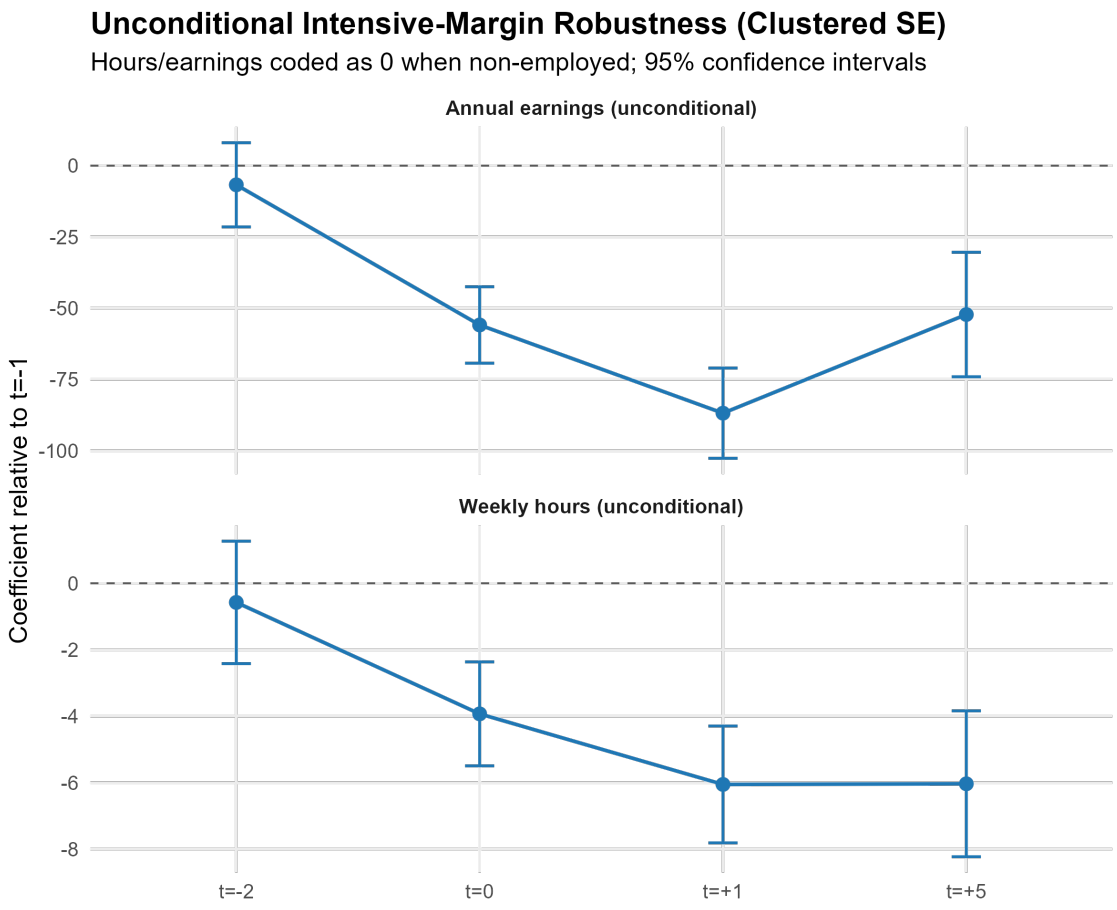
	$t = -2$	$t = 0$	$t = +5$
Pooled	-0.012 ($p = 0.70$)	-0.132 ($p < 0.001$)	-0.028 ($p = 0.38$)
Early ($\leq$2012)	$+0.020$ ($p = 0.63$)	-0.131 ($p < 0.001$)	-0.020 ($p = 0.66$)
Late ($\geq$2013)	-0.063 ($p = 0.19$)	-0.135 ($p = 0.002$)	-0.025 ($p = 0.66$)
No-COVID	-0.004 ($p = 0.91$)	-0.145 ($p < 0.001$)	-0.028 ($p = 0.40$)

The birth-year break is virtually identical across cohorts and unaffected by excluding 2020–2022.

Backup: IPW and Attrition

- Observability at $t = +5$ is weakly selective on baseline employment ($p = 0.045$)
- IPW reweighting:
  - Near lead: unchanged
  - Birth-year break: unchanged (-0.136, $p < 0.001$)
  - $t = +5$: shifts to -0.057 ($p = 0.076$) - more negative
- Balanced panel ($t = -2$ to $t = +1$, N=662, 2,648 obs): core coefficients preserved

Backup: Unconditional Outcomes



Standard event-study coefficients relative to $t = -1$. Year FE, robust SEs. Coding 0 for non-employed preserves the post-birth declines and near-lead null.

Backup: Missingness Sensitivity (Profile)

Table: Missingness profile for mechanism covariates in the childbirth-margin risk set (employed at $t = -1$, observed at $t = 0$).

Covariate missingness	N	Share of risk set	Exit rate at $t = 0$
Contract type missing	36	0.109	0.444
Firm size missing	16	0.048	0.750
Either missing	48	0.145	0.500

Notes: Risk-set size is $N = 330$ women. Exit is defined as non-employment at $t = 0$ among women employed at $t = -1$.

Backup: Missingness Sensitivity (Mechanism)

Table: Childbirth-margin mechanism sensitivity to missing-data handling.

Specification	N	Exits	OR Non-regular	p	OR Small firm	p
Complete-case	282	100	7.505	0.000	1.548	0.126
Missing-indicator	330	124	7.476	0.000	1.289	0.345
High-information	237	83	7.393	0.000	1.860	0.047

Notes: High-information sample requires non-missing contract type, firm size, commute indicator, and husband's overwork indicator at $t = -1$. Missing-indicator specification includes missing dummies for contract and firm size.

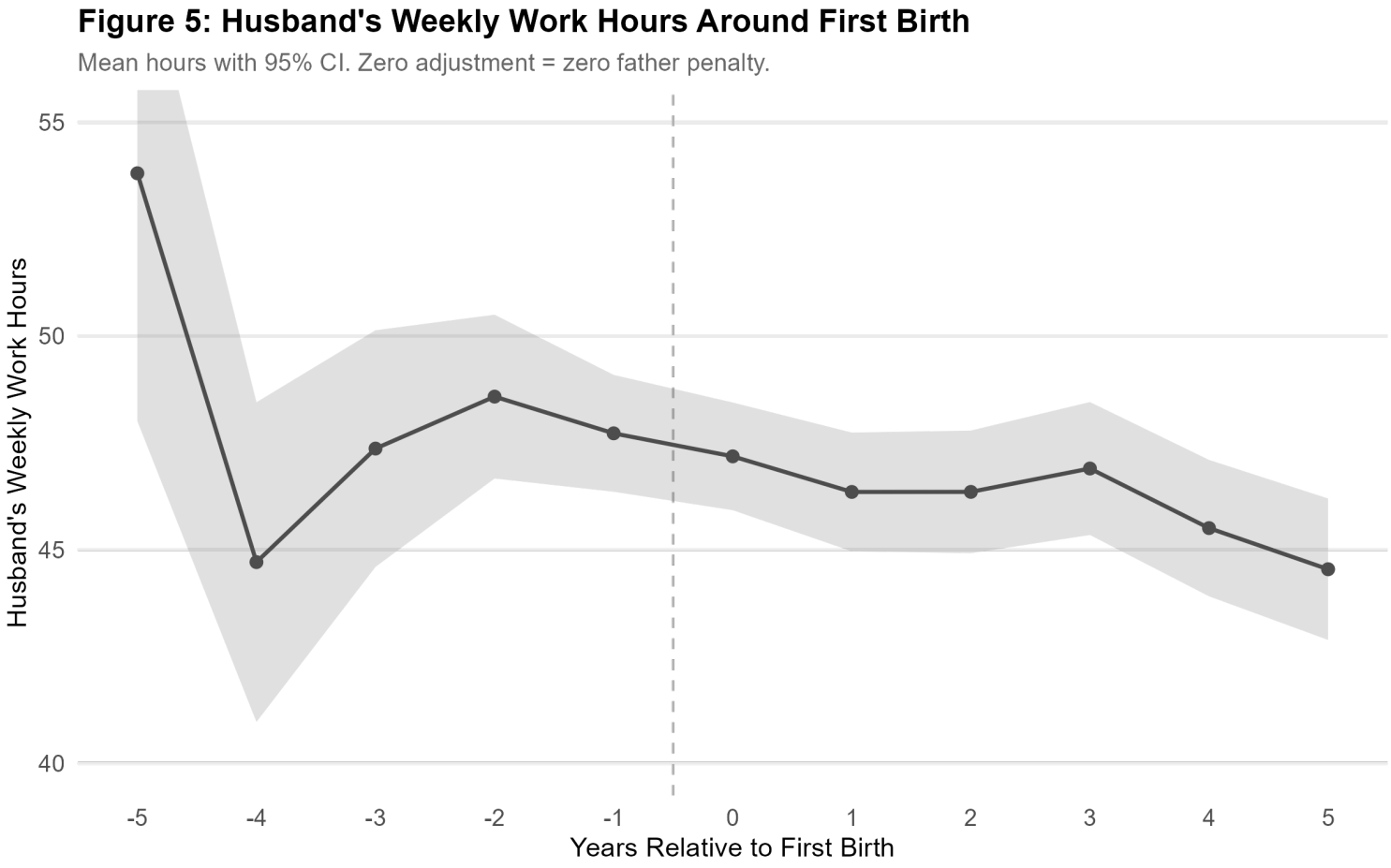
Non-regular OR stable: 7.50, 7.48, 7.39 across specifications.

Backup: Pre-Birth Mechanism ($t = -2$, 13 events)

	OR	p	95% CI	Method
Non-regular				
Standard	6.41	0.019	-	MLE
Firth	5.32	0.009	-	Penalised
Bootstrap	-	0.008	[1.68, 19.66]	2,000
Small firm				
Standard	4.14	0.025	-	MLE
Firth	3.84	0.021	-	Penalised
Bootstrap	-	0.038	[1.10, 18.34]	2,000

Jackknife: non-regular OR above 1 in all 185 replications (range: 5.69–12.82).

Backup: Father-Side Evidence



Flat profile. Childcare ratio 6.3:1 at $t = +1$, 5.3:1 at $t = +5$.
Only 3/1,183 husbands (0.3%) took childcare leave.

Backup: Training Grants Usage

Event time	N	Skills training (%)	Grants used (%)
-2	253	35.2	2.2
-1	475	31.2	2.3
0	568	8.6	1.0
$+1$	594	10.6	1.3
$+3$	493	15.6	0.3
$+5$	437	12.8	0.0

Training drops from $\sim$35% to 8.6% at birth, recovers to $\sim$13%.
Government grants: $<$3% at every horizon. System is invisible.

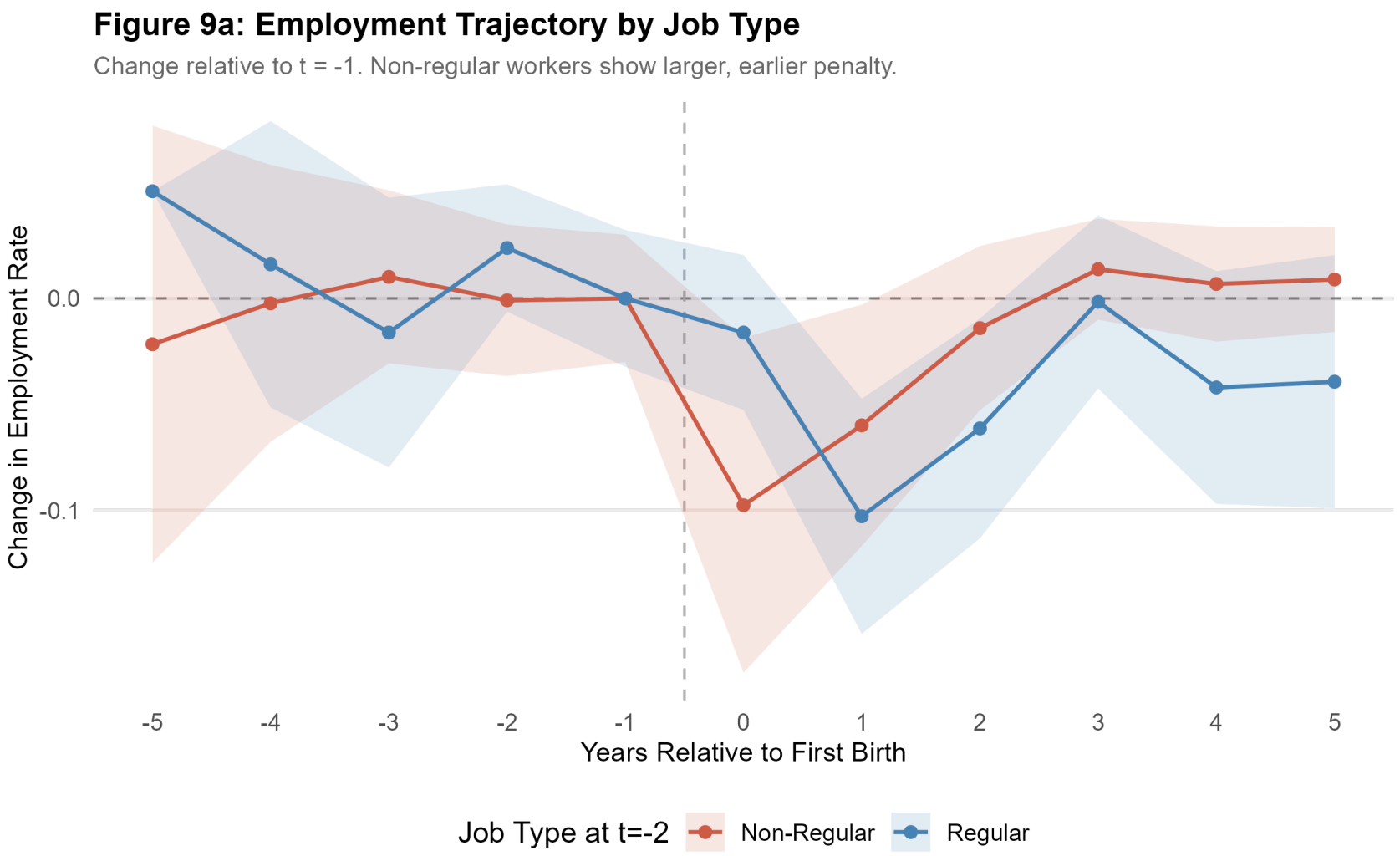
Backup: KHPS/JHPS Split

Table: Survey-split stability check for key employment event-study coefficients (year fixed effects).

Sample	Event time	Estimate	SE	p-value	N obs	N women
Pooled	$t = -2$	-0.012	0.031	0.703	5,212	662
Pooled	$t = +0$	-0.132	0.027	0.000	5,212	662
Pooled	$t = +5$	-0.028	0.032	0.384	5,212	662
KHPS	$t = -2$	0.002	0.039	0.966	3,323	394
KHPS	$t = +0$	-0.134	0.036	0.000	3,323	394
KHPS	$t = +5$	0.010	0.040	0.796	3,323	394
JHPS	$t = -2$	-0.027	0.049	0.576	1,889	268
JHPS	$t = +0$	-0.136	0.044	0.002	1,889	268
JHPS	$t = +5$	-0.126	0.057	0.027	1,889	268

Near lead and birth-year break are stable across surveys. Long horizon diverges.

Backup: Heterogeneity by Contract Type



Non-regular: -9.8 pp at $t = 0$. Regular: -1.6 pp. Five-fold difference.

Backup: Earnings Bunching by Event Time

Backup: Geographic Non-Variation

- Metropolitan (Kanto/Kinki, N=355): -13.6 pp at $t = 0$
- Non-metropolitan (N=307): -12.9 pp at $t = 0$
- Near lead null in both subsamples

The dual labour market operates nationwide.

Backup: Leave-Recoding Sensitivity

Table: Leave-as-employment sensitivity at childbirth ($t = 0$).

Specification	Coef. at $t = 0$	SE	p-value	Reclassified at $t = 0$
Original (hours > 0)	-0.1319	0.0275	<0.001	0
Leave-adjusted	-0.0988	0.0276	<0.001	22

Notes: Sensitivity recodes women as employed at $t = 0$ if they are coded non-working but report leave take-up near birth. The coefficient remains strongly negative after reclassification.

Reclassifying 22 leave-takers as employed attenuates $t = 0$ from -0.132 to -0.099; remains strongly significant.

Backup: Full Stability Map

Stable	Sensitive	Interpretive
Near lead null (-0.012, $p = 0.70$)	Long horizon (composition-sensitive)	Marriage-stage sequencing
Birth-year drop (-0.132, $p < 0.001$)	Far leads reflect changing support	Childcare-logistics narrative
Non-regular OR at $t=0$ (7.50, $p < 0.001$)	Pre-birth mechanism (13 events)	Household-structure channels
Cohort-split stable (early $\approx$ late $\approx$ no-COVID)	Industry splits selection-sensitive	Cross-country framing
Bunching at 103 man-yen (47.8%)	Hours/earnings cond. on employment	Training non-uptake causes

Additional checks: placebo, IPW, balanced panel, Lee bounds, clustered SEs, KHPS/JHPS split, missingness sensitivity, jackknife.